TabAI — Complete .env Setup Guide

Step-by-step: from zero accounts to a working backend. Total time: ~20 minutes.

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1 Supabase (PostgreSQL Database)

~5 min — Free tier: 2 projects, 500 MB database, unlimited API requests

Step 1.1 — Create account

1. Go to `https://supabase.com`
2. Click **"Start your project"**
3. Sign up with **GitHub** (fastest) or email
4. Verify your email if prompted

Step 1.2 — Create a new project

1. Click **"New Project"**
2. Choose your organization (auto-created on signup)
3. Fill in:
   - **Project name:** `tabai`
   - **Database password:** click **"Generate a password"** — *COPY THIS PASSWORD NOW, you won't see it again!*
   - **Region:** pick the closest to you (e.g. `eu-central-1` for Europe)
   - **Plan:** Free
4. Click **"Create new project"**
5. Wait ~2 minutes for provisioning

> ⚠️ **Save the database password!**
>
> Write it down immediately. If you lose it, go to Settings → Database → Reset database password.

Step 1.3 — Get your credentials

API keys — go to **Settings** → **API** (left sidebar):

What to copy	Goes into .env as

Project URL — looks like `https://abcxyz123.supabase.co`	`SUPABASE_URL`
anon / public key — starts with `eyJ...`	`SUPABASE_ANON_KEY`
service_role / secret key — starts with `eyJ...`	`SUPABASE_SERVICE_KEY`

Database URL — go to **Settings** → **Database**:

1. Scroll to **"Connection string"** section
2. Select the **"URI"** tab
3. Copy it — looks like: `postgresql://postgres:[YOUR-PASSWORD]@db.abcxyz123.supabase.co:5432/postgres`
4. **IMPORTANT:** Change the prefix from `postgresql://` to `postgresql+asyncpg://`

📝 **Your .env values so far:**

```
DATABASE_URL=postgresql+asyncpg://postgres:YOUR_PASSWORD@db.YOUR_REF.supabase.co:5432/post
SUPABASE_URL=https://YOUR_REF.supabase.co
SUPABASE_ANON_KEY=eyJ...
SUPABASE_SERVICE_KEY=eyJ...
```

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2 Cloudflare R2 (Audio File Storage)

~8 min — Free tier: 10 GB storage, 10 million reads/month, zero egress fees

Step 2.1 — Create Cloudflare account

1. Go to `https://dash.cloudflare.com/sign-up`
2. Sign up with email + password
3. Verify your email
4. You do NOT need to add a domain or website — skip any onboarding prompts

Step 2.2 — Enable R2 and create bucket

1. In the left sidebar, click **"R2 Object Storage"**
2. You may need to enter payment details (credit card) — **you won't be charged** on the free tier, but Cloudflare requires a card on file for R2
3. Click **"Create bucket"**
4. Name: `tabai-audio`
5. Location: **Automatic** (or pick your region)
6. Click **"Create bucket"**

Step 2.3 — Get your Account ID

1. Look at the URL in your browser — it looks like: `https://dash.cloudflare.com/abc123def456/r2`
2. That bold part is your **Account ID**
3. Or: go to any R2 page → right sidebar shows **"Account ID"**

Step 2.4 — Create an R2 API token

1. Go to **R2 Overview** page
2. Click **"Manage R2 API Tokens"** (top right area)
3. Click **"Create API token"**
4. Configure:
   - **Token name:** `tabai-backend`
   - **Permissions:** `Object Read & Write`
   - **Specify bucket(s):** select `tabai-audio`
   - **TTL:** leave default or set to never expire
5. Click **"Create API Token"**
6. **COPY BOTH VALUES IMMEDIATELY** — they are shown only once:
   - **Access Key ID** — looks like a short string
   - **Secret Access Key** — longer string

> ⚠️ **Copy the secret immediately!**
>
> Cloudflare shows the Secret Access Key only once. If you lose it, delete the token and create a new one.

> 📝 **Your .env values so far:**

```
R2_ACCOUNT_ID=your-32-char-account-id
R2_ACCESS_KEY_ID=your-access-key
R2_SECRET_ACCESS_KEY=your-secret-key
R2_BUCKET_NAME=tabai-audio
```

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3 Modal (GPU Serverless Workers)

~5 min — Free tier: $30/month credits (enough for ~100 transcriptions)

Step 3.1 — Create Modal account

1. Go to `https://modal.com`
2. Click **"Sign up"**
3. Sign in with **GitHub** (required — Modal uses GitHub auth)
4. You'll be dropped into the Modal dashboard

Step 3.2 — Authenticate CLI

In your terminal (with the backend venv activated):

```
cd backend
source venv/bin/activate
modal setup
```

1. This opens your browser
2. Click **"Approve"** to link your terminal to Modal
3. You'll see: `✓ Authentication successful`

Step 3.3 — Create a Modal secret

The GPU worker needs access to your database and R2. Create a secret bundle:

```
modal secret create tabai-secrets \
  DATABASE_URL="postgresql+asyncpg://postgres:PASS@db.REF.supabase.co:5432/postgres" \
  R2_ACCOUNT_ID="your-account-id" \
  R2_ACCESS_KEY_ID="your-access-key" \
  R2_SECRET_ACCESS_KEY="your-secret-key" \
  R2_BUCKET_NAME="tabai-audio"
```

> ✅ **Use the SAME values**
>
> you already put in your .env file for DATABASE_URL, R2_ACCOUNT_ID, R2_ACCESS_KEY_ID, and R2_SECRET_ACCESS_KEY.

Verify it was created:

```
modal secret list
```

You should see `tabai-secrets` in the list.

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4 Generate SECRET_KEY

~10 sec

Run this in your terminal:

```
openssl rand -hex 32
```

Copy the output (64-character hex string) and paste it as `SECRET_KEY` in your .env.

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5 Optional: Redis & Sentry

Variable	What to do
`REDIS_URL`	Keep default `redis://localhost:6379`. Only needed if you run Docker Compose or install Redis locally. Not required for basic operation.
`SENTRY_DSN`	Leave empty. Optional error monitoring. Sign up at `sentry.io` if you want it later.

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6 Final .env File

Your completed `backend/.env` should look like this:

```
# === Database (Supabase) ===
DATABASE_URL=postgresql+asyncpg://postgres:MyP4ssw0rd@db.abcxyz123.supabase.co:5432/postgres
SUPABASE_URL=https://abcxyz123.supabase.co
SUPABASE_ANON_KEY=eyJhbGciOiJIUzI1NiIsInR5cCI6IkpXVCJ9.eyJpc3MiOi...
SUPABASE_SERVICE_KEY=eyJhbGciOiJIUzI1NiIsInR5cCI6IkpXVCJ9.eyJpc3MiOi...

# === Storage (Cloudflare R2) ===
R2_ACCOUNT_ID=a1b2c3d4e5f6a1b2c3d4e5f6a1b2c3d4
R2_ACCESS_KEY_ID=abcdef1234567890abcdef12
R2_SECRET_ACCESS_KEY=abcdef1234567890abcdef1234567890abcdef12345678
R2_BUCKET_NAME=tabai-audio

# === App ===
REDIS_URL=redis://localhost:6379
SECRET_KEY=147d471d53857a37544d4a822b4c8fa71778ae9029f2531686575dbbec262f2c
ENVIRONMENT=development
CORS_ORIGINS=http://localhost:3000
SENTRY_DSN=
```

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7 Verify Everything Works

Terminal 1 — Start backend

```
cd backend
source venv/bin/activate
uvicorn api.main:app --reload --port 8000
```

Expected: `Development mode: database tables created.` + `Uvicorn running on http://127.0.0.1:8000`

Terminal 2 — Start frontend

```
cd frontend
npm run dev
```

Expected: `Ready in X.Xs` at `http://localhost:3000`

Terminal 3 — Deploy Modal worker

```
cd backend
source venv/bin/activate
modal serve workers/transcription.py
```

Expected: `Serving app "tabai-transcription"`

Verify checklist

Check	How	Expected
Health	Open `http://localhost:8000/health`	`{"status":"ok","version":"0.1.0"}`
API docs	Open `http://localhost:8000/docs`	Swagger UI with /api/jobs endpoints
Frontend	Open `http://localhost:3000`	TabAI upload page with drop zone
Upload test	Drop a short MP3 file	Progress bar moves through all steps → tab displayed

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Troubleshooting

Problem	Fix
`nodename nor servname provided`	DATABASE_URL has wrong Supabase project ref — double check in Supabase dashboard
`password authentication failed`	Wrong DB password — reset it in Supabase Settings → Database
`403 Forbidden` on R2 upload	R2 token doesn't have write permissions, or wrong bucket name
`SignatureDoesNotMatch` on R2	R2_SECRET_ACCESS_KEY is wrong — create a new API token
`Modal dispatch failed`	Run `modal setup` again, or check `modal secret list` shows `tabai-secrets`
Frontend shows 404	EMFILE error on macOS — run `ulimit -n 10240` before `npm run dev`